

Lec 42 Style GAN 2

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Transcript

[00:00] Hello everyone, welcome back to the lecture series on fundamentals of generative AI and large language models. In today's session, we will discuss about style GAN 2. In my previous session, we have discussed about style GAN which is a variation of GAN architecture. Now there is some modification that scientists have done to style GAN. We will learn about that one which is our topic of discussion today and it is named as style GAN 2.

[00:33] So the contents for today's session goes like limitations of style GAN origin of water droplet like artifacts the architectural changes in style GAN 2 weight modulation and demodulation origin of phase artifact solution techniques for phase artifacts in style GAN 2. skip connection in generator and residual connection in the discriminator. So we have seen in our previous lecture how style GAN works. Now what is the need for style GAN 2? When style GAN was giving us realistic good images generated images what is the requirement that motivated style GAN 2. So let us have a look into that one. Is there any limitation of style GAN for which this style GAN 2 came into picture? We have to look into

[01:30] that. So if we see here in this uh uh picture, this is taken from the original paper where where style GAN 2 was introduced. We will see this is a picture of a

human face and at this point it may not be visible here in the uh image. But if we look at the feature map there is a kind of artifact a kind of a you know uh water droplet kind of a thing that is present here. Similarly for this image also at this point there is a kind of an artifact here. So if you see images and their feature maps there is some kind of artifact it is coming everywhere. Okay. So that is one thing which uh the researchers found that there is some structural thing or architectural thing in style GAN which is consistently

[02:30] giving these kind of artifacts. Okay. So that was the reason they wanted to have a look into the architecture and change it in such a way that these artifacts can be removed. Okay. So there are characteristic artifacts that were generated by style GAN. So these are the images that were generated by style GAN and there are two types of artifacts which um you know came into um people discovered. One is water droplet like artifact. Another one is phase artifact. Okay. Now we will have a look into how the architecture of style GAN was changed. So we don't get any of these artifacts.

[03:17] Style GAN 2 was developed to remove these artifacts. That is the main reason for the uh you know style GAN coming into picture. The paper in which style GAN 2 was introduced is from the same group which introduced style GAN and his team and the name of the paper is analyzing and improving the image quality of style GAN. So style GAN used to give good quality images with controlled styles. But in style GAN 2 these artifacts which were coming as a structural problem in style GAN or structural limitation in style GAN that has been modified in style GAN 2 so that artifacts don't come into picture. So we'll have a quick review on style GAN 1. What was the architecture of style GAN one? We know that in style GAN the

[04:12] lat the input from the latent space it used to go through the mapping network and the latent space uh used to be a um entangled latent space where all styles were entangled and after going through the mapping network it used to get transformed into another space which is a disentangled uh latent space and then that used to go through the learned a fine transformation which used to give style as uh you know parameters like the scaling and the shift and these used to go as input to the add in operation and u in the synthesis network the synthesis used to start from a learned tensor constant tensor of $4 + 4 + 512$ size and that was the input and when uh the style parameters were introduced into ADA in

[05:11] the ADA in had two steps of operation. One is to normalize the feature maps and another one is to modulate with these style parameters. And then there was a progressive growing of the resolution. So this block is having $4 + 4$ resolution. Next block is having $8 + 8$ resolution. So this was the architecture of the synthesis network. And we have seen that there were at every step there was some noise injection that was happening to bring uh bring in some stochastic u variation in the images that are produced. Okay. So this was the architecture of style GAN. Now I have taken this

image from once again the paper uh stylebased generator architecture for generative adversarial network that is the paper where style

[06:00] got introduced. Okay. Now if we have a look into the artifacts that I was speaking about in my previous slide, these artifacts begin to appear in the resolution 64×64 resolution. Before that it was not visible but from 64×64 resolution uh it started appearing and it progressively increased or grown in strength as the resolutions increases. So we have resolutions in style gan till 1024×1024 . So all throughout the process from 64×64 to $1,024 \times 1,024$ there was a um increase in the strength of these artifacts. Okay.

[06:45] Now we have to see what is the origin of these water droplet like artifacts. Okay. Now as I was telling in normal uh style GAN the the traditional style GAN that we have studied before the uh adine injects tiles at each layer and when adine operates on a feature map it normalizes the mean and variance of the feature map. So this is known to us. So it normalizes it first normalizes and then it modulates with the style parameters that are coming from the aine transformation.

[07:20] But this step of normalizing the mean and variance of the feature map interfered with the signal statistics. Now when I tell signal statistics what exactly I mean for any statistics in the signal here I'll mean mean the mean and the variance of each channel of the feature map. Okay. So when the feature map gets normalized the statistics the signal statistics gets interfered and what happens due to this one there is uh some that is the reason some artifacts come into picture. Now why that is coming because mean and variance of each channel of the image they carry the semantic information. So they have a lot of information about the features that are present in the feature map. Now before addin's operation the feature

[08:19] feature mean was meaningful that means it conveyed the semantic information of the image. Similarly the feature variance also conveyed meaningful information about the image. Now after addin's operation what we see is this is the normalization part and then this multiply this gets multiplied by γ si which is the scaling parameter that comes from the a fine transformation and this is the shift parameter γ vi that comes that also comes from as a part of your output of your learn and uh a fine transformation. Now what happens here is when ADA in normalizes the mean and the variance generator also learns to exploit the normalization statistics. When I tell this one what I mean exactly is when I am when the mean and the variance are

[09:21] normalized in ADA in what exactly they are trying to do? they are trying to remove the semantic information or the style information that is present in that feature map and then inject the new styles. Now generate what is generator's work? The generator have to produce images by which it can fool the discriminator. Okay. So in this case generator learns how to exploit the normalization statistics. So it knows that if I change some of the values in the mean and the variance, if I can manipulate

the mean and the variance, then I can control the style in the feature map. Okay. So that is that is from there the origin of water droplet like artifact come into picture. So what happens is the generator creates one

[10:15] extremely large or strong localized peak in an image. Okay, like a single pixel or a very tiny region that goes to a very high value and that dominates the statistics of the entire instance normalization computes. So what exactly I want to say uh here is suppose Suppose I have a feature map with like 0 1 2 uh 7 something like this. Now if the generator changes it like you know say 0 100 27 the generator spikes one of these values to a very high value. Now when I am taking the mean and the variance of this you know vector and compared to this one the entire mean and the variance will be dominated by this value. Correct? Because think of a class of u uh students where all the students have got

[11:31] 0 to 100 marks and one student out of all those students scored 10,000 points. Then what happens the ex the extreme outlier here which is 10,000 it controls the scale at which each of the students score belongs. Now so the mean the variance the everything will now but will change. Okay so this has a dominating um uh this becomes a dominating factor in the mean and variance calculation. So the generator uses this trick to fool the discriminator. It spikes one of the uh values and then the mean and variance gets completely determined by that one speak. So, so it is entirely dependent or determined that by that one spike and this spike manifests blob-like structure or you know uh water droplet kind of artifacts okay in the

[12:29] generated images. So when we know this change in the statistics in ADA in is the main reason for the water droplet kind of artifacts then what is the solution that is proposed by the researchers or the scientists. So what scientists did is they removed ADA in normalization completely and introduced something called as the weight modulation and demodulation. Now what is that one? What do I mean by weight modulation and demodulation?

[13:08] Instead of modifying the feature maps directly the styles now modify the convolution weights. So we have convolution operations in inside the uh generator network. So in this case in style GAN 2 the feature maps are not modified. What is getting modified here with the style parameter is the convolution weights. Now how it happens and that is called as the weight modulation. So if we see this expression $w'_{jk} = s_i w_{jk}$. So this is my original weight of the convolution matrix and this is my modulated weight. Okay. w' prime is my modulated weight and s_i is the style scaling factor that is present in the iat layer. w_{ijk} these are the convolution weights. Okay. And i JK I represents the input channel. J

[14:14] represents the output channel and K represent the location of the weights inside the convolution filter. Okay. So what we are seeing in the weight modulation what is happening? We are modulating the weights of the convolution uh filter. Okay.

Instead of modifying the feature maps. And what is demodulation? In style GAN 2 we have weight modulation plus demodulation. So after modulation what happens is some channels become extremely large. Okay. Because we are multiplying by the factor uh S_i . Okay.

[14:48] So which is the styling uh scaling uh parameter from u_m style scaling parameter in coming from the a fine transformation. Now if some channel values become too high what happens is this causes instability and that in to reduce that instability we undergo the process of demodulation. What is it that is normalize the magnitude of the weights. So the demodulation equation goes like this. $W'_{ijk} = \frac{W_{ijk}}{\sqrt{\sum_i \sum_k W_{ijk}^2 + \epsilon}}$. Now we are not going into the derivation of this one. However we need to understand what each and every term here you know is u what each and every term here means. what is the significance of the presence of these

[15:49] terms? So this is basically W'_{jk} . This is basically your modulated weight and u_m so this is basically your modulated weight and then there is a summation over i and k . So it is the summation it adds over all input channels I and uh all all input channels I and over all the kernel positions K . Okay. And then there is a square term of $\sum_i \sum_k W_{ijk}^2$. So W'_{jk} is your modulated weight weight. Why are we taking the square?

[16:29] Because if the weights are positive and negative, there is a chance of getting cancelled. So to avoid the cancellation of positive and negative weights, this squared term is taken. Okay. And there is a square root. This square root restored the normalized energy back into the original scale. Okay. So that is the reason for putting the square root term and there is a term ϵ . We know usually ϵ is a very small number which is added to avoid division by zero. Okay. So this is the equation for demodulation of the weights. And why do we demodulate them? because during modulation it is possible that some of the channels may become very high or large in the numbers. So in order to get rid of that one this demodulation

[17:19] step is done. Now if we have to see let us go through the weight modulation and demodulation pipeline. So we have the latent vector z . Then we have the mapping network that goes to the style mapping network gives output the style vector W . This goes to the fine transformation and we get after a fine transformation that style parameter the scaling parameter it is used in the weight modulation to to modulate the convolution u_m convolution filters weights. Okay.

[17:59] And then convolution happens with these modulated weights. And the final feature map gets demodulated or the weight demodulation is happening after the convolution operation happens. Okay. And then there is noise injection. We all know a noise injection is done to bring in some stochastic variation in the images. So any image that is generated image one and image two if they are generated they will not

be exactly same they there will be some stoastic variation which is given by this noise injection and then we get the feature map. So this is the modulation demodulation pipeline okay in style GAN 2. Now this is a picture once again I have taken from the original uh paper where style GAN 2 was introduced and if you

[18:54] see this is the image and this is the feature map and we don't see any artifacts here. This is the image of a human face. This is the image of a car. This is the image of a horse or a man you know riding the horse. And these are the feature maps associated but there is no artifacts that are present here. Okay. So weight modulation then demodulation it helped to get rid of the artifacts that were coming while images were getting generated in using style GAN architecture. Weight modulation and demodulation was introduced in style GAN 2 for removing these artifacts and here it is clearly demonstrated.

[19:36] Now we come to another um uh type of artifact which we call as the phase artifact. Now what is that phase artifact? Let us have a look into this picture. This is the face a human face. It is facing the camera straight facing the camera. And you can see the teeth is aligned at with the center of the camera. Now look at the second image here. The face gets little bit rotated but this but still the center of the teeth is still aligned with the center of the camera there. Usually what happens when we uh turn our face the entire positions of our teeth the nose the eyes they get changed but in this case that is not happening. It is still fixed. Okay. So here also you see there is a movement in the uh faces uh

[20:36] position but still the center is aligned the center of the teeth is aligned to the center of the camera and so you can see there is an offset here. Okay. So these kind of artifacts are called as you know uh phase artifacts. So as I was telling when the face rotates teeth and all other features of the face should rotate but in this case what happens is they still remains aligned to the camera's position camera center probably and then suddenly jumps okay so this kind of artifact is called as phase artifact or blob and in uh you know why it is called it is called as phase artifact because in signal processing phase refers to spatial alignment or positional consistency of patterns. So any artifact that is coming into picture

[21:28] because of positional inconsistency that artifact is named as phase artifact. Okay. Now we have to look into what is the origin of this phase artifact. The main reason for this uh phase artifact coming into picture is the progressive growing during the training phase of style GAN. So what do you mean by progressive growing? We have seen in style GAN that the training initially starts from $4 + 4$ resolution. Then it goes to $8 + 8$ $16 + 16$ $32 + 32$ $64 + 64$ cross 64 it goes on till $1,24 + 1,024$. So this is called as the progressive progressive growing of the resolution.

[22:12] Now what happens is during the training it starts in original style GAN training begins at a very low resolution which is $4 + 4$. Now $4 + 4$ resolution after the

first you know uh layer it gives uh you know an output okay and that feature map output that intermediate resolution temporarily acts as the final output $4 + 4$ or $8 + 8$ or $16 + 16$ these are very low resolution images they don't have enough of spatial uh they don't have the ability for higher spatial resolution. But what happens here is during the training [snorts] when we are training the first layer say $4 + 4$ or $8 + 8$ the layers are forced to generate very fine details too early. these fine details the resolution doesn't have that capability

[23:17] but still they are forced to generate the fine details and that temporarily acts as the final output. So whatever details that we are getting that becomes temporarily that acts as a final input. What I'm telling will be little bit more clear when we talk about high frequency and uh regions and low frequency regions. So let us have a look into this picture. Okay. So okay now there is here is an image of a [clears throat] leaf. Okay. If we see the center suppose I take these pixel at the center the s the adjacent center adjacent pixel also is having similar color. So the value change from this pixel to this pixel is not much. If there is a pixel shift, if this pixel goes here or this pixel goes

[24:22] here, there is a pixel shift, then it will not show up as a distortion because the values are almost similar and the image is smooth here. The colors are smooth. Small changes, one pixel shift will not distort the image. Okay, these are called as the low frequency regions. Whereas have a look into this edge, the edge of the leaf. Now if I look into this last tip there is a sharp you see this pixel or if you take maybe a portion of this pixel and this pixel.

[25:16] If I shift this pixel here, the shape of the leaf will get distorted because the pixel values here are very much different. The green value and the whitish value at the background they are very very there is a sharp change in the pixel values. So whenever we are coming to some sharp edge like this say and if I want to shift this pixel here then what happens the the same thing shifts here. This becomes the part of the leaf and then what happens this entire portion gets distorted.

[25:58] Okay. So these are called as the high frequency region. Okay. Now why we are talking about this high frequency and low frequency region we will understand in this slide. So progressive growing causes high frequency problem. What exactly is meant by this one? Suppose we are looking during the training phase. So the network is training at $64 + 64$. Now why do I take $64 + 64$? If you remember in my first or second slide, we have told the artifact starts coming visible from this $64 + 64$ uh resolution and then it starts becoming stronger and stronger with the progressive growing of the resolution. Okay, progressive improve uh yeah progressive growing of the resolution. Now network let us take we are in the $64 + 64$ resolution and the

[26:54] network is getting trained at 64×64 okay uh resolution then what happens here is this layer must already create some realistic images. So whatever be its

resolution capability uh still it has to create some realistic images. Okay. Now the $64 + 64$ they don't have enough spatial precision. Maybe one pixel is covering a large image portion. So from where it will give the finer details. It is not possible to give the finer details. Okay. Now what happens here is in that case whatever details it is giving the features become position dependent and that is one of the main reasons for this misalignment of features as and when we go into the progressive growing uh phase. So [clears throat] high frequency

[27:55] features are very sensitive to location. As I was telling if there is one pixel shift also there is distortion there is flickering effects there is inconsistency in the spatial structures. Okay as I was discussing with the leaf image. Now if there is uh uh you know if the high frequency reg if we consider the high frequency regions and we've see that little bit of pixel shift or little bit of disalignment gives you a distorted image that becomes a major problem in progressive growing because in each and every step of progressive growing we expect slight misalignment.

[28:39] ment. Okay, misalignment of the position with respect to the previous one. Now we'll just have a look how that problem is coming here. So we'll have a small example which will give us an idea what exactly I'm telling that as we go grow in the pro as the uh resolutions keep increasing there is a problem with the alignment spatial alignment. [clears throat] Suppose we have two pixels pixel 0 and pixel 1 and the values of these two pixels here it is under pixel 0 it is zero and this is one. So basically it is the black and white. So if we see the edge between the black and the white we can expect it exactly at the center. Exactly at the center. Okay. Now so then we are upsampling this one. So here instead of two pixels we will get

[29:34] four four pixels in the same same you know region pixel 0 1 2 3. Now suppose the values are like this 0.3.6.1. So this is complete black. This is complete white and in between there is gray portion. Okay. So in this case where should the uh edge remain. So we have to see in this case there is a question mark. Where will the age be? Will the age be here? Will the age be here? Will the age be here? Where exactly? Because this entire portion is gray. Yeah, there is a variation in the shade of the gray but there is some ambiguity.

[30:23] So there is ambiguity regarding the position of the edge. Okay. So and think of the situation here itself there is an ambiguity. It is not it may not exactly match with the edge here. Okay. So it may shift if there are multiple layers of upsampling. These misalignments keep on accumulating and at the end of $1024 + 1024$ resolution what you get that may be quite a bit of shift of alignment with your actual image. So this misalignment comes with upsampling.

[31:13] Misalignment of special uh features comes with upsampling and this becomes a very important or significant problem in case of highfrequency features because these high frequency features are very much position dependent or posi-

tion sensitive. Okay. So that is the entire reason why we are getting artifacts like these phase artifacts come into picture in style GAN. So we have seen that the phase artifacts come because of the progressive growing. So what is the solution that is proposed by the researchers to overcome this problem? It is they remove the progressive growing completely and there are three things that they have done. one is kept the architecture fixed in style GAN 2 used skip connection in the generator and

[32:05] residual connection in the discriminator. So we will have a look one by one into each of them and see how they have been able to you know overcome this phase artifact problem. So we'll first look into keeping the architecture fixed. So during training phase what happens is in original style GAN in the stage one only $4 + 4$ resolution images are getting trained and in stage two say 8×8 new 8×8 block is inserted in stage three new 16×16 block is inserted. So every layer one new block is getting inserted. So the architecture changes continuously and this keeps on going till we reach the 1024×1024 resolution. This is the case in original style GAN. Whereas in style GAN 2 what they have done is

[33:00] from the first training iteration only they have kept 4×4 8×8 the entire network was fixed. So there was no new layers that were added later. So the architecture is fixed. That is what has been told here. Keep the architecture fixed. Don't keep on adding new layers in the architecture during the training phase. So that is one of the uh you know uh changes that they have done in the architecture in style GAN 2 compared to original style GAN. Then what the second thing that they have done is skip connections. They have introduced skip connections in the generator. Now what exactly happens here in original style GAN generator that is without the step skip connection we had 4×4 feature maps then it goes to 8×8 cross uh 8

[33:51] feature maps and it keeps on going till 1024×1024 feature maps we get resolution feature maps we get and finally we get an RGB image. Okay. So what happens here in this case is so here it was like every feature map every resolution had to wait to contribute to the final RGB all across all these uh across all these layers. But in style GAN 2 what they did is with skip connection they have done 4×4 feature map when it is generated a portion of that goes you know they keep for the RGB. So the final RGB lay image that is coming out it is getting inputs from all the layers. The difference between original style GAN and style GAN 2 is here the entire process of progressive entire process of going through the various

[34:51] layers of convolution was done and finally the RGB uh image was produced. Whereas in style GAN 2 what happens is after 4×4 resolution processing image processing we get the uh feature map 4×4 feature map and portion of it is kept for 2×2 RGB and similarly 8×8 feature map a portion goes to RGB and 16×16 feature map a portion goes to RGB. So what happens is your final image is getting RGB contributions from all the feature maps in you know uh once the this feature map is produced

it is getting so what happens here is without skip connection the early layers have to wait through many layers to affect the output. So the early layer this 4×4 layer have to wait till it it reaches 1024×1024 to

[35:46] contribute to the final RGB. Okay. And what happens is all throughout these uh you know u what layers of convolution the spatial shifts get accumulated. Okay. And what happens is this whatever comes in the final final RGB the spatial shift is already there. Whereas with skip connection the earlier layers directly influence the final um image. So a portion directly goes to the RGB and multiscale RGB outputs are progressively combined. So this is very very important because from every scale 8×8 4×4 8×8 16×16 from every scale we are getting the RGB outputs which are combined together to give you the final image.

[36:31] Okay. So that is the concept of a skip connections in the generator and this helps in better gradient flow improved training stability. So overall the artifact the phase artifact can be you know uh or the uh spatial um shifts can be reduced or can we can get rid of that one using this skip connections in the generator. Then the third step that they have taken scientists have taken to get out of this artifact problem or the uh phase artifact problem is residual connections in the discriminator. So skip connections in the generator and residual connections in the discriminator. So what is the core idea here? The phase shifts are basically coming from spatial misalignment or texture drifting or the features are uh

[37:24] you know tied to the pixels that is the main reason for phase shift artifacts coming into picture. So the discriminate what now this residual connections are coming in the discriminator part. So what is the function of the discriminator? The discriminator should tell the generator or force the generator to produce properly aligned image features so that this artifact doesn't come into picture. Now when the progressive growing stage or the progressive growing is completely removed in style GAN 2, the discriminator is given the full resolution image from the beginning itself. Okay. So the the image that is coming to the discriminator at the first instance itself is 1024×1024 . Now this is a large image and the discriminator's

[38:21] network becomes deep in order to process this image. Now we'll just have a visual representation of this original style GAN. We were having these upsamplings like this from 4×4 to 1024×1024 . So the initial part the discriminator initially handled very small image. So this is 4×4 that is handled and you know handling a small image of 4×4 resolution doesn't require a deep neural network. It requires a shallow network that is sufficient for it. When it comes to 1024×1024 or maybe a layer above that one the discriminator in this positions they are handling larger images and it requires deeper network. Okay that was in original style GAN. So when progressive growing is removed the discriminator from the initial step

[39:18] itself have to process the large image which is 1×1 2×2 4×4 1×1 24×24 . Now

processing this large image by the discriminator it requires a lot of downsampling. Why it requires a lot of downsampling? What is the function of the discriminator? The discriminator's function is to look into the generated image and real image and then tell which one is generated. For that one it has to look into the features. It has to look into all the details that are present in the image. If it starts you know uh from a very high resolution one it it it will be memory intensive computational intensive. So all throughout it cannot you know it it will be too um yeah memory intensive uh calculations will be there. So what it does is it down

[40:12] samples from $1024 + 1024$ all the way down to $4 + 4$. Okay. So each stage contains convolution layers making the network deep in from the beginning itself. So so many layers it has to you know uh process and the problem that comes with deep network is gradient may vanish that uh the training may become unstable optimization becomes difficult because it is a deep network. Now how residual connections in the discriminator how it helps to uh overcome this problem? We know residual connections work like this. So if we have an input here x and it under goes some convolution layers and we get an output from here and uh if output from the convolutions the learn transformation is f_x then this input uh

[41:04] also comes and gets added here and the output that we get here is f_x plus x . Okay. So what exactly we are trying to say here is whatever information is available in the first layer let us not destroy it let us add it with the learned transformation. Okay so if the original input is already having useful information which is there in our $1024 + 1024$ image the network should not destroy it. So the information is getting added here and residual connection ensures the original information is added to new learned refinements without undergoing you know all these layers. So directly through the uh residual connection information from the first layer can get added to this layer to the output.

[41:56] Without residual parts what happened the repeated convolutions as you we were telling before also may distort the feature locations. So the spatial misalignment gets added or accumulated over the multiple uh uh convolution layers that through which the image is getting processed or the feature maps are getting processed and this weakens the initial uh alignment uh information that was there in the original feature map. So that's how the discriminator overcomes the problem of deep network using residual connections. The generator uses skip connections and the architecture is kept fixed in style GAN 2. And with these three steps scientists were able to come out of the or overcome the problem of phase artifacts. So with this one I

[42:51] come to the conclusion of um uh today's session. Before I conclude let me quickly summarize what we have studied today. So we have started with what is the need for style GAN 2. Tile GAN itself was giving us good realistic images, high resolution images. But it was seen that even though artifacts were not seen in the final image but if we look into the feature maps from $64 + 64$ resolution every feature

map you know in the increasing progressive growth of the resolution uh the artifacts were present and there are two types of artifacts water droplet kind of artifacts and phase artifacts and we have seen what is the origin of the water droplet artifacts what is the origin of the phase artifact and what is the solution

[43:39] that has been taken for uh you know for the for removing the water droplet kind of artifact and in uh what is the solution that has been taken in the phase artifact removal. So we have seen in the uh water droplet kind of artifact they have removed the adda inoperation completely and it is replaced by weight modulation and demodulation due to which the convolution uh convolution weights were getting modulated and deodulated and with this one they have been able to come out of that uh water droplet kind of artifact. Phase artifact comes from the misalignment of uh spatial features over and and it becomes very prominent for high frequency features which are very very sensitive to position and the

[44:34] solution that was um suggested by scientists were three things. Keeping the architect remove progressive growth growing completely keeping the architecture fixed from the beginning. So in style gen 2 we have the network fixed starting from 8 cross 4 + 4 to 1,024 + 124 1,024 and they have introduced skip connections in generators and residual connections in discriminators. Skip connection in generators helped to um solve the problem of this misalignment of spatial misalignment. Whereas the uh residual connection in the generator in the discriminator it helped to overcome the problems of a deep network in the discriminator. So with all these things scientists were able to overcome the problem of artifacts in highly realistic

[45:36] images that are produced by style GAN and uh in style GAN 2 basically and uh with this one I come to the end of uh today's session uh in the next session uh we will discuss about diffusion models which is another um complete another topic uh for imaging generation. Okay. So we will take up diffusion model in my next session and till that time keep learning, happy learning and thank you all.